8 PERIODIC MAINTENANCE

8.1 Overview

This chapter describes the maintenance and inspection procedures required to maintain the seal checker in optimum operating condition.

CAUTION

- **Before performing maintenance/inspection work of the system, turn OFF the power switch, lock the power switch and keep the key by the worker. Otherwise, other worker may operate the system resulting in injury.**

8.2 Daily Inspection

8.2.1 Pre-Startup Inspection

Before starting production, perform the inspection procedure described below.

Table 8-1 Table of Inspection at Commencement of Work

Inspection Item	Description	Measures
System and around the system	• Verify that tool or unnecessary object for sorting operation is not left on the top or around the system.	Remove.
	• Each belt is not deformed or worn out.	Replace the conveyor belt.
	• Each sensor functions properly.	Clean the sensor lens.
	• Each conveyor has no play. • There is no unusual sound generated from conveyor, motor and gear box.	In the case of unusual sound or meandering, consult the distributor you purchased or our company.
	• Reject operation is performed properly.	Consult the distributor you purchased or our company.

8.2.2 Inspection during Production

CAUTION

- **When performing inspection during production, do not be too close to the equipment, or touch the conveyors or the moving parts.**

Table 8-2 Table of Inspection during Operation

Inspection Item	Description	Measures
Belt condition.	Visually check that each conveyor belt has no damage, deterioration or meandering.	Ask the distributor you purchased or our company for replacement of conveyor belt.
Noise generation	Verify that each conveyor belt has no meandering	In the case of unusual sound or meandering, consult the distributor you purchased or our company.
	Verify that each conveyor belt is not tensioned too much.	
	Verify that no unusual sound is generated from the rotating part.	

8.3 Maintenance and Inspection Procedures

8.3.1 Printer Paper Replacement

NOTE

- The contents of this item are applicable only when the machine is delivered in the simple substance.

When the message "printer: no paper" appears in the lower part of the RCU display, replenish the roll by following the procedure below.

1. Press down on the printer cover to open the cover.

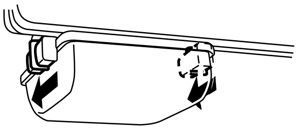

Fig.8-1 Printer External View

2. Pull the fixed lever toward, and press down the printer unit.

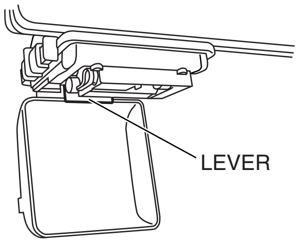

Fig.8-2 Fixed Lever

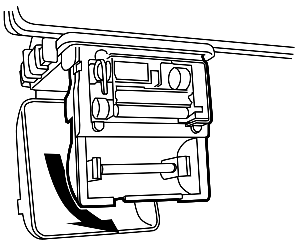

Fig.8-3 Printer Unit

3. Raise the printer head lever.

   ▶The printer head will stand upright.

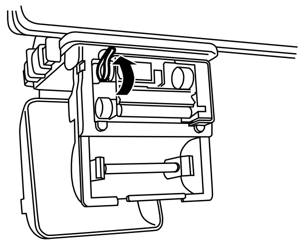

Fig.8-4 Printer Internal View

4. Press open the roll holder clips and remove the empty roll core.
5. Press open the roll holder clips and remove the core of the old roll.
6. Insert a new roll into the clips in the direction shown on the right.

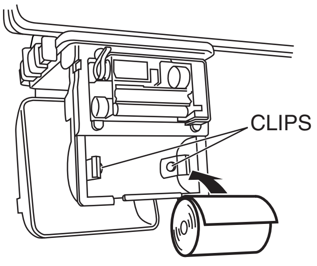

Fig.8-5 Paper Roll Loading

7. Insert the end of the paper into the printer.
8. Feed the paper and pull out about 1cm from the printer head.

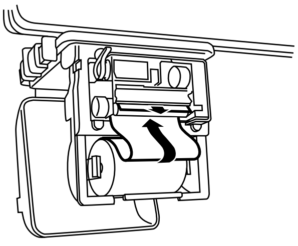

Fig.8-6 Printer Paper Threading

9. Press the printer head lever.

   ▶The printer head will return to the lowered position.

10. Press up the printer unit.
11. Shut the printer cover.

   ▶Paper roll replacement will be complete.

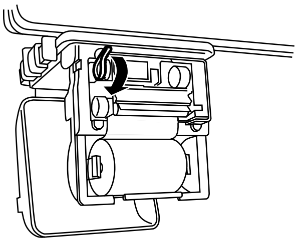

Fig.8-7 Printer Internal View

8.3.2 Operation check of photo sensor

This system is provided with two photo sensors.
This photo sensor has the following functions.

WARNING

- **When working with the main power switch ON, tag out that the system is being adjusted to prevent other persons from operating the system.**

1. A broken bag by entanglement into the packer is detected.
2. Seal check conveyor inlet:
   Products carried into the seal checker are detected.

NOTE

Verify that S (green) is lit all the time while the power is ON.
Also verify that L (red) goes ON/OFF according to the existence of product to be inspected.

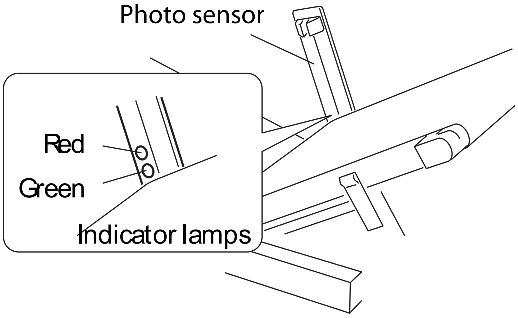

Fig.8-8 Photo sensor View(Option)

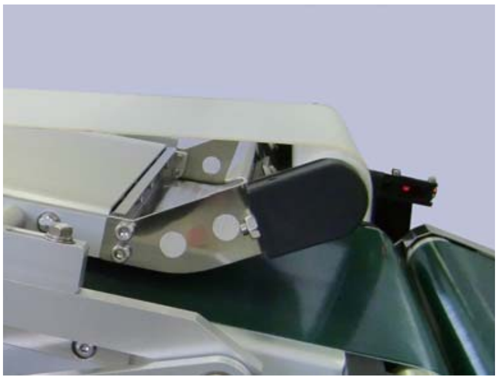

Fig.8-9 Photo sensor View(Option)

8.3.3 Timing belt tension adjustment and visual check for scratch

The proper tension requires a gap of approximately 10 mm between the driving motor gear and the timing belt at the operation.

1. Turn off the main power of equipment.
2. Check that the conveyor belt and the conditioning brush do not operate.

Fig.8-10 The brush conditioner

3. Remove the screws securing the timing belt cover.

Fig.8-11 Remove the timing belt cover

4. Hold the handle of brush conditioner with both hands and move it to the down side sliding.

CAUTION

- **Be sure to use both hands when moving the brush conditioner. Otherwise, the driving part may catch your fingers.**

Fig.8-12 Sliding the brush conditioner

5. Raise the timing belt cover to remove it.

6. The proper tension (approximately 10 mm of the gap between the gear and the belt) is approximately 10 to 15 mm of deflection made when the center of timing belt is lightly pressed by finger.
   Check that there is no scratch or cut on the timing belt. If there is any scratch or cut, contact ISHIDA service Center.

Fig.8-13 Check the belt tension

How to adjust when the tension is not enough

1. Lightly loosen the four screws securing the driving motor and move the motor to strain the tension.

   Attach the timing belt cover to tighten the securing screws.

   Hold the handle of conditioning unit with both hands to slide and move to the upper side.

8.3.4 Height Compensation of Seal Check Conveyor

Perform the height compensation once a month.

1. Turn on the main power of equipment.
2. Press the [H COMPE] Key.

▶The height compensation mode starts. When the message "Put the compensation block" appears, put the compensation block on the Infeed Conveyor straightforward.

Fig.8-14 Press the [H COMPE] Key

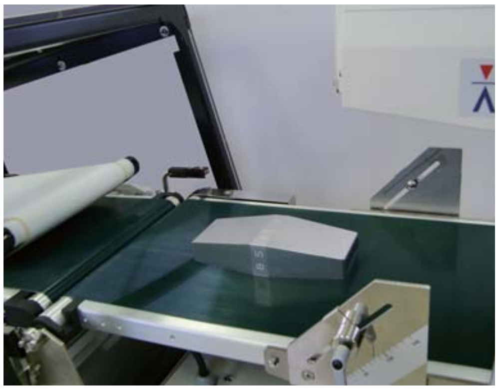

Fig.8-15 Place the compensation block

3. Press the [OK] key.

▶The conveyor starts and the compensation block is fed to the inspection part.

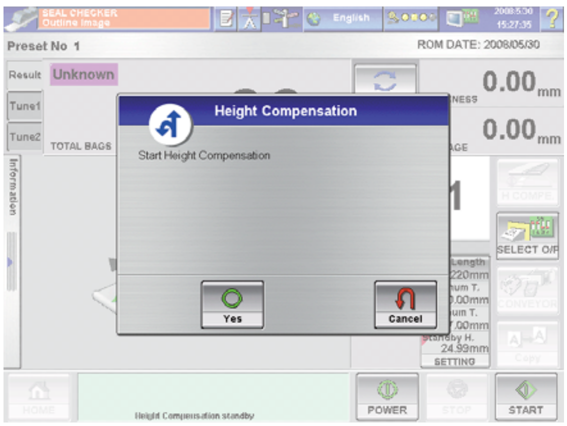

Fig.8-16 Press the [OK] key

4. The compensation completion message appears. Then check that the status is POWER OFF on the screen.
5. Open the cover.
   Raise the Seal Check Conveyor with both hands to remove the compensation block.
   The height compensation is completed.

CAUTION

- **Ensure that the Seal Check Conveyor does not catch your fingers.**

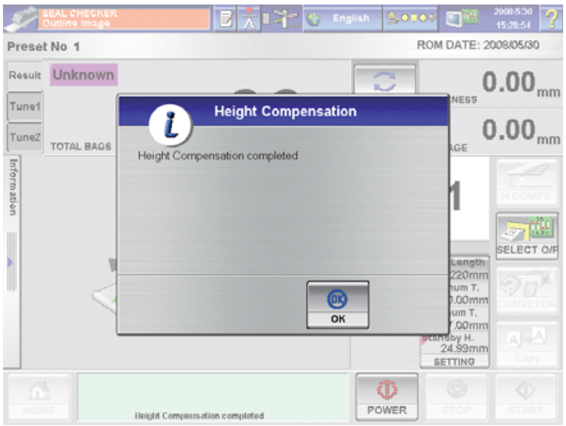

Fig.8-17 POWER OFF screen

8.3.5 Timing belt tension adjustment

Adjust the timing belt tension for smooth power transmission and improvement of durable years of driving parts.

CAUTION

- **Do not provide the timing belt with too much tension.**
  It will decrease the durability of timing belt and durable years of bearing.

- Timing belt tension adjustment

Move the motor mount and adjust the timing belt tension so that the deflection should be appropriate when lightly pushed the central part of timing belt with a finger.

NOTE

The position of drive timing belt is shown. (Refer to Fig.8-8 Timing Belt Position Chart.)
Adjust the following drive timing belts in the same manner:

- J-belt conveyor drive timing belt
- Infeed conveyor drive timing belt
- Lower seal check conveyor drive timing belt
- Upper seal check conveyor drive timing belt
- Reject conveyor drive timing belt
- Brush Conditioner drive timing belt

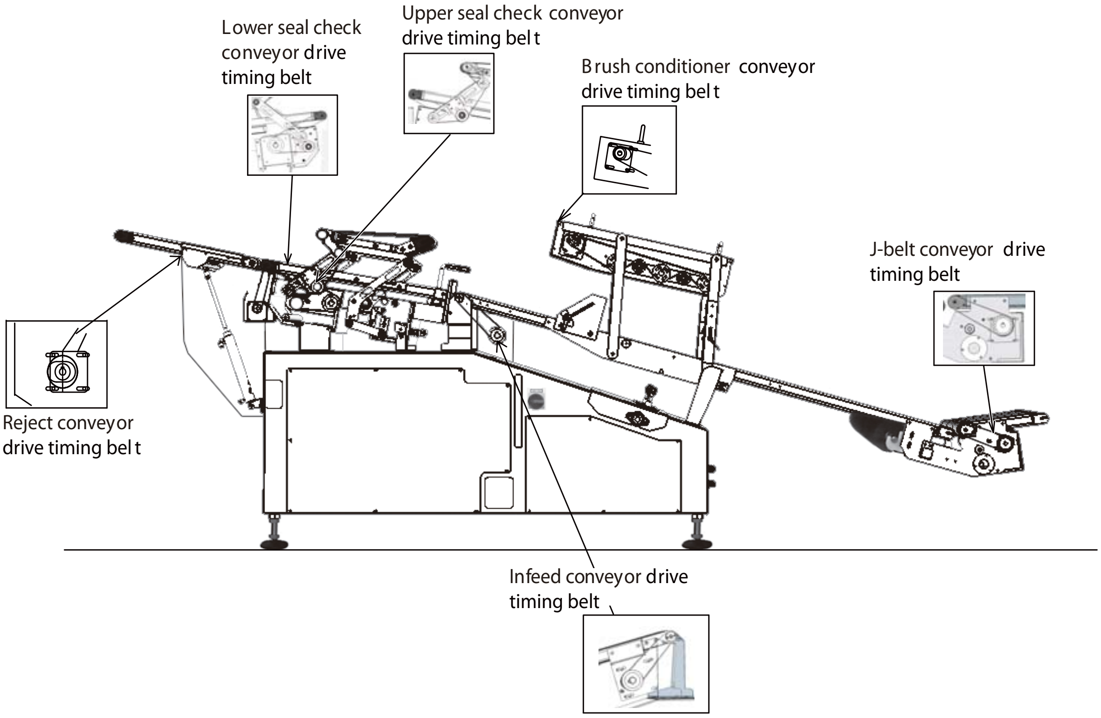

Fig.8-18 Timing Belt Position Chart

8.3.6 Inspection / adjustment of conveyor belt

CAUTION

- **Do not apply too much tension to the belt.**
  It will decrease the durability of bearing and belt.
- **Apply the tension equally to the left and right of the belt in crosswise direction.**
  Unequal tension may cause meandering of the belt.

For smooth transfer, check if the belt is slipping due to too much or too small tension of conveyor belt.
If the belt slips, apply the tension not to slip with the tension adjusting bolt (both sides).

NOTE

Adjust the following belts:
(Refer to Fig.8-11 Conveyor Tension Adjusting Bolt Position Chart.)

- J-belt conveyor
- Infeed conveyor
- Upper seal check conveyor
- Lower seal check conveyor
- Reject conveyor
- Brush Conditioner

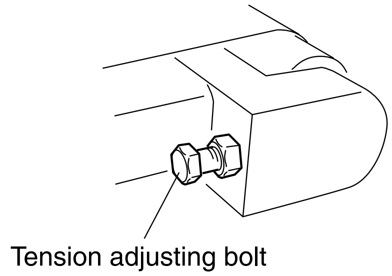

Fig.8-19 Conveyor Tension Adjusting Bolt

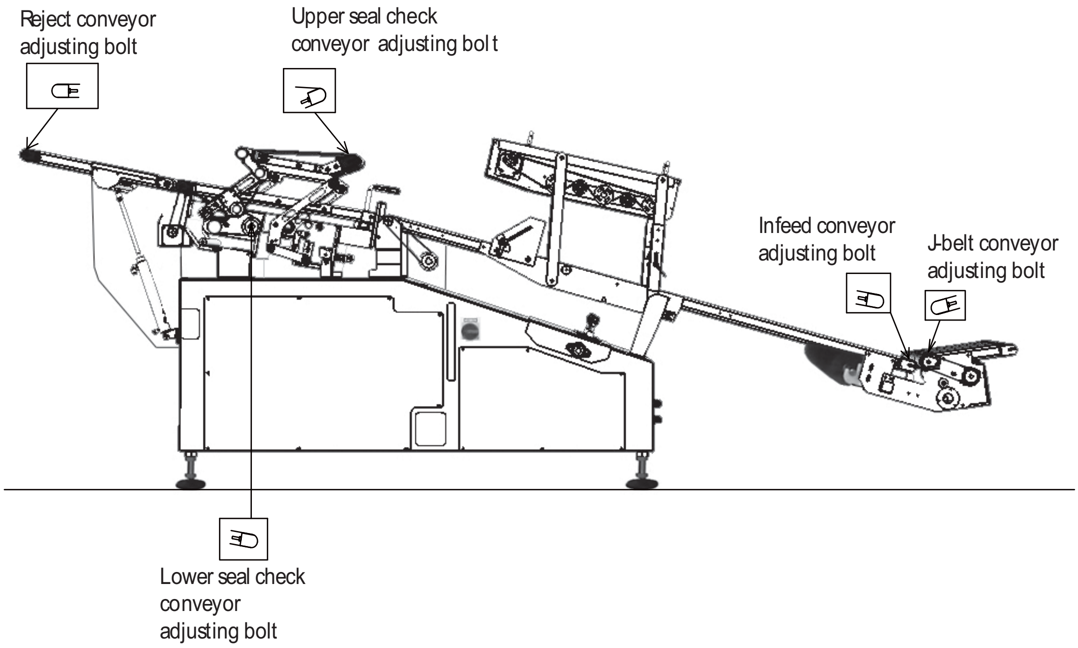

Fig.8-20 Conveyor Tension Adjusting Bolt Position Chart